Free balancing equations worksheet worksheet 22

Maths | 8th grade | Algebra

1. If 3x + 2 = 11, what is the value of x?

A) 3
B) 2
C) 7
D) 11

2. Solve for y: 5(y - 2) = 20.

A) 4
B) 6
C) 8
D) 10

3. What is the solution to the equation 4(x + 1) = 2(x + 6)?

A) x = 2
B) x = 3
C) x = 4
D) x = 5

4. Which value of x satisfies the equation 2x^2 - 8 = 0?

A) 2
B) -2
C) 0
D) 4

5. If 7 + 3x = 2x + 10, what is x?

A) 2
B) 3
C) 4
D) 5

6. Solve for z: z/4 + 3 = 7.

A) 12
B) 16
C) 20
D) 24

7. What is the solution to the equation 9 - 2x = 3?

A) 2
B) 3
C) -3
D) 4

8. If 5x - 1 = 2x + 11, what is the value of x?

A) 4
B) 5

C) 6
D) 7

9. Solve for x: 2(x - 3) = x + 2.

A) 1
B) 2
C) 3
D) 4

10. What value of n satisfies 3n + 5 = 2n - 4?

A) 1
B) -1
C) -9
D) 9

Answer Key

1. A
Subtract 2 from both sides to get 3x = 9, then divide by 3 to find x = 3.

2. C
Distribute 5 to get 5y - 10 = 20, add 10 to both sides to get 5y = 30, then divide by 5 to find y = 6.

3. A
Expand both sides to get 4x + 4 = 2x + 12. Subtract 2x from both sides and subtract 4 from both sides to find 2x = 8, so x = 4.

4. A
Add 8 to both sides to get 2x^2 = 8, then divide by 2 to find x^2 = 4, so x = 2 or x = -2.

5. B
Subtract 2x and 7 from both sides to get x = 3.

6. B
Subtract 3 from both sides to get z/4 = 4, then multiply by 4 to find z = 16.

7. A
Subtract 9 from both sides to get -2x = -6, then divide by -2 to find x = 3.

8. C
Subtract 2x from both sides and add 1 to both sides to get 3x = 12, then divide by 3 to find x = 4.

9. C
Distribute to get 2x - 6 = x + 2, subtract x and add 6 to both sides to find x = 8.

10. C
Subtract 2n and 5 from both sides to get n = -9.